BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Biology (Tenth, 2022, Lahore Board) (Objective Type) (Group - I)

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Specific combination of genes in an individual is known as :

(A) Phenotype
(B) Genotype
(C) Dominant
(D) Homozygous

Q2. The enzyme that is used to cut gene of interest is:

(A) Lipase
(B) Amylase
(C) Restriction endo nucleases
(D) Ligase

Q3. Mature bone cells are called:

(A) Cartilage
(B) Osteocytes
(C) Chondrocytes
(D) Endoskeleton

Q4. Symbiosis in which both partners get benefit is:

(A) Parasitism
(B) Predation
(C) Competition
(D) Mutualism

Q5. Which of these addictive drugs are also used as pain killer:

(A) Narcotics
(B) Sedatives
(C) Hallucinogens
(D) All can be used

Q6. What type of blood vessels surrounds the alveoli:

(A) Artery
(B) Arteriole
(C) Capillary
(D) Vein

Q7. Which part of the female reproductive system receive eggs cells from the ovary:

(A) Fallopian tube
(B) Uterus
(C) Vagina
(D) Cervix

Q8. Which organ is responsible for filtering the blood:

(A) Intestine
(B) Brain
(C) Stomach
(D) Kidney

Q9. Hydrogen bonds present between cytosine and guanine are :

(A) Two
(B) Four
(C) Three
(D) One

Q10. Rhizopus reproduces asexually by:

(A) Binary fission
(B) Budding
(C) Spore formation
(D) Endospore formation

Q11. Cochlea is present in :

(A) Internal ear
(B) External ear
(C) Middle ear
(D) External and middle ear

Q12. The base of food chain is always a:

(A) Plant
(B) Sun
(C) Ocean
(D) Moon

Biology (Tenth, 2022, Lahore Board) (Subjective Type) (Group - I)

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

QPART - I.

QQ.2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) What are alveoli? What is their function?

(ii) Explain aerobic respiration.

(iii) Differentiate between glottis and epiglottis.

(iv) Define excretion.

(v) What is the function of ureter?

(vi) Write down name of different types of neurons.

(vii) Write a short note on midbrain.

(viii) What is hippocampus?

QQ.3. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Define hinge joints Give an example.

(ii) Differentiate between osteoporosis and arthritis.

(iii) Define double fertilization.

(iv) Why do gardeners use the methods of cutting and grafting?

(v) What Is meant by homologous chromosomes?

(vi) What are discontinuous variations?

(vii) Write any two nitrogenous bases of DNA.

(viii) Compare the breeds and cultivars.

QQ.4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) What are omnivores? Give an example.

(ii) What are the reasons of competition among organisms of ecosystem?

(iii) Write the name of two types of pollutants.

(iv) Which are environment friendly fuels to reduce air pollution?

(v) Write uses of ethanol.

(vi) Write two objectives of genetic engineering.

(vii) What is drug? Write the name of its two types.

(viii) Differentiate between bactericidal and bacteriostatic antibiotics.

Biology (Tenth, 2022, Lahore Board) (PART-II) (Group - I)

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

> Note: Attempt any TWO questions.

QQ.5(a). (Marks 5) What is nephron? Explain its structure in detail.

QQ.5(b). (Marks 4) Explain in detail the function of skeletal muscles.

QQ.6(a). Describe any two disorders of nervous system. [marks obscured in the source scan]

QQ.6(b). (Marks 4) Describe the two mechanisms of animal fertilization.

QQ.7(a). (Marks 5) Write basic steps in genetic engineering in detail.

QQ.7(b). (Marks 4) Write a detailed note on narcotics and marijuana.

GROUP – II

Biology (Tenth, 2022, Lahore Board) (Objective Type) (Group - II)

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The thickness of Biosphere is about:

(A) 40 km (B) 20 km

(C) 10 km (D) 30 km

Q2. Which part of embryo in seed give rise to plant roots:

(A) Cotyledon (B) Radical

(C) Plumule (D) Testa

Q3. Separate out painkiller drug:

(A) Analgesics (B) Antiseptics

(C) Antibiotics (D) Sedatives

Q4. The length of trachea is about _____:

(A) 10 cm (B) 12 cm

(C) 14 cm (D) 16 cm

Q5. Which of these is an anti-viral protein:

(A) Urokinase (B) Thymosin

(C) Insulin (D) Interferon

Q6. Which organ is responsible for filtering the blood:

(A) Intestine (B) Brain

(C) Stomach (D) Kidney

Q7. Dengue fever is a ------- infection :

(A) Bacterial (B) Fungal

(C) Viral (D) Algal

Q8. Cartilage is a type of tissues:

(A) Muscle (B) Smooth

(C) Connective (D) Cardiac

Q9. All of these are hormones except:

(A) Insulin (B) Thyroxin

(C) Glucagon (D) Pepsinogen

Q10. The example of co-dominance is:

(A) Blood Group A (B) Blood Group B

(C) Blood Group AB (D) Blood Group C

Q11. Material containing sperms and fluid is called:

(A) Semen (B) Uterus

(C) Cervix (D) Vas-deferens

Q12. The number of hydrogen bonds between adenine and thymine is:

(A) 2 (B) 3

(C) 4 (D) 1

Biology (Tenth, 2022, Lahore Board) (Subjective Type) (Group - II)

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

QPART - I.

QQ.2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Define inspiration.

(ii) What is a glottis?

(iii) Define bronchitis?

(iv) Write the size of kidney?

(v) Define lithotripsy.

(vi) Define coordinator.

(vii) Write the function of occipital lobe.

(viii) Why owl cannot see during the day time?

QQ.3. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Define bone and give its examples.

(ii) What is tendon?

(iii) What is meant by cloning?

(iv) Define mutation.

(v) What is gene flow?

(vi) Differentiate between genotype and phenotype.

(vii) What is nucleosome?

(viii) Enlist two sources of variations.

QQ.4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Write the effects of acid rain.

(ii) Define smog.

(iii) Write effects of eutrophication.

(iv) Write the difference between renewable and non-renewable natural resources.

(v) Write dairy products of fermentation.

(vi) Define single cell protein.

(vii) Define vaccines.

(viii) Define pharmacology and distinguish it from pharmacy

Biology (Tenth, 2022, Lahore Board) (PART-II) (Group - II)

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

QQ.5(a). (Marks 2,2,1) Describe the functioning of kidney.

QQ.5(b). (Marks 1,1,1,1) What is cartilage? Describe its types.

QQ.6(a). (Marks 5) Explain in detail the internal structure of human ear.

QQ.6(b). (Marks 4) Describe the structure of seed.

QQ.7(a). (Marks 1,2,2) What is fermenter? Describe its using ways.

QQ.7(b). (Marks 1,3) What are antibiotics? Write not on their three major groups.